

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

CHEMISTRY

PAPER 2 Theory

4024/2

2 hours

SPECIMEN PAPER

Additional materials:

Mathematical tables/ Electronic calculator (optional)

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on all separate answer paper used.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer any **four** questions.

Write your answers on the separate answer paper provided

At the end of the examination, fasten any separate answer paper used securely to the question paper.

Enter the numbers of **Section B** questions you have answered in the grid.

All essential working must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

A copy of the Periodic Table is printed on page 14.

FOR EXAMINER'S USE

Section A

Section B

TOTAL

This specimen paper consists of 14 printed pages.

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Section A

Answer all questions form this section.

Write your answers in the spaces provided on the question paper.

1. (a) **Table 1** is an incomplete table showing pollutants from the Contact process.

Table 1

pollutant	method used to minimise the pollutant
unreacted SO ₂	using CaCO ₃
(i)	
(ii)	

Complete **Table 1** by stating the other two pollutants and the methods used to minimise them.

[4]

- (b) (i) State the role of yeast in the fermentation of glucose

.....
.....

[1]

- (ii) Give **one** other condition necessary for the fermentation process.

.....
.....

[1]

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3

- (iii) Ethanol produced in **b (i)** is dilute.

Describe and explain how the dilute ethanol solution is concentrated

.....
 [2]

2. (a) A white powder, **M**, dissolves in water to form a solution with a pH of 3.5.

- (i) State the effect of solution **M** on blue litmus paper.

.....
 [1]

- (ii) When solution **M** is added to a solid, **Q**, carbon dioxide gas is formed.

Describe, with a reason, the chemical composition of **Q**.

.....
 [2]

- (b) Two unknown solutions containing magnesium and zinc ions were analysed.

Describe the chemical tests that can be done to show the presence of each ion..

test(s)

.....
 [3]

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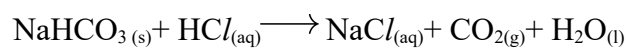
observation(s)

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[2]

3. (a) A mass of 3.50 g of sodium hydrogen carbonate reacted completely with 40 cm³ of 1.0 mol/dm³ hydrochloric acid.

The equation for the reaction is shown:



Write the ionic equation of the reaction

[1]

- (b) Calculate the

(i) the number of moles of HCl that reacted,

(ii) the volume of CO₂ produced.

[5]

- (c) Deduce with a reason the limiting reagent

.....
 [2]

4. (a) (i) The equation shows the reaction between sodium thiosulphate and dilute hydrochloric acid.



State what the symbol (s) represents.

..... [1]

- (ii) Suggest any **three** ways of measuring the rate of the reaction.

1

2

3

..... [3]

- (iii) State and explain any **two** factors that can increase the rate of the reaction.

factor 1

explanation.....

factor 2

explanation.....

[4]

5. (a) **Fig. 8.1** shows three electric cells that were used to compare reactivity of three metals A, B and C.

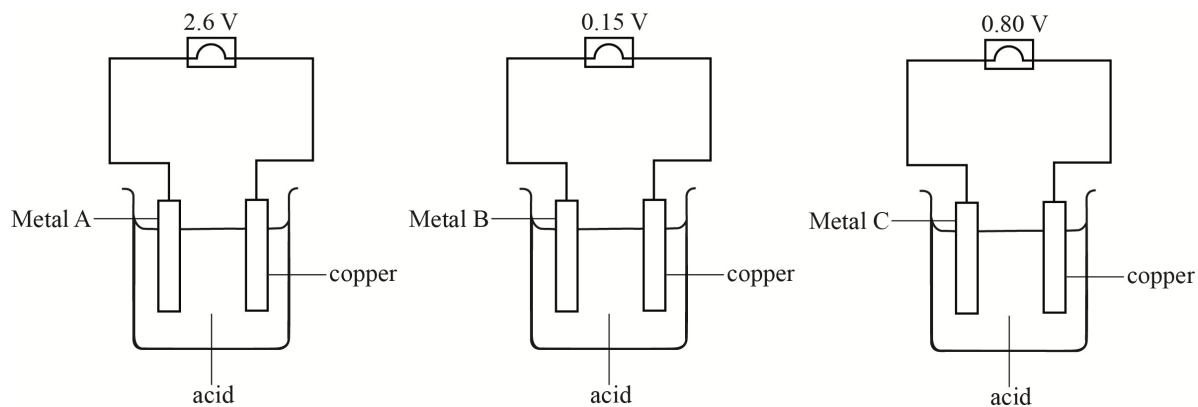


Fig.8.1

- (i) Explain why metals A, B and C are all more reactive than copper.

.....
 [1]

- (ii) Arrange the metals A, B and C in order of decreasing reactivity (starting with the most reactive).

..... [1]

- (iii) Explain the trend in (ii).

.....
 [1]

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- (b) (i) State and explain the trend in the reactivity of group (VII) elements.

trend

explanation

[3]

- (ii) Deduce with reason(s) the physical state of astatine at room temperature.

state

reason(s)

[2]

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Section B

Answer any four questions from this section.

Write your answers on a separate answer paper provided.

6. (a) (i) The electronic configurations of three elements, **X**, **Y** and **Z** are shown.

X: 2. 8.2

Y: 2. 8. 7

Z: 2. 6

State with a reason the period to which **Y** belongs. [2]

- (ii) Write the chemical formula of the compound formed when **X** and **Y** chemically combine. [2]

- (iii) Determine the formula of the ion formed by **Z**. [1]

- (b) Table 6 shows physical properties of three substances **A**, **B** and **C**.

Table 6

substance	melting point/ $^{\circ}\text{C}$	electrical conductivity when molten	electrical conductivity when solid
A	1084	conducts	conducts
B	808	conducts	does not conduct
C	119	does not conduct	does not conduct

Identify with reasons the type of bonding in

- A**
- B**,
- C**.

[5]

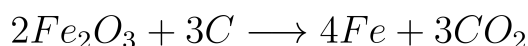
- (c) A mass of 10.00 g of impure calcium carbonate reacted with excess hydrochloric acid to produce 140 cm³ of carbon dioxide at r.t.p.

Calculate the

1. mass of CaCO₃ in the sample,
2. percentage purity of the sample.

[5]

7. (a) (i) Name the ore from which iron is obtained. [1]
- (ii) Write a balanced chemical equation for the reaction of iron oxide with carbon monoxide. [2]
- (iii) Describe how slag is formed during the extraction of iron. [2]
- (b) Explain why water stored in metal iron containers usually has a brown colour. [2]
- (c) Iron oxide is reduced by carbon as shown in the equation:



- (i) Calculate the mass of iron oxide needed to produce 14 000 kg of iron. [2]
- (ii) Calculate the amount of ore required to produce 14 000 kg of iron given that 40 % of the ore are impurities. [1]
- (d) (i) Describe any **two** human activities that increase soil acidity. [2]
- (ii) State how the high levels of acidity can be reduced. [1]
- (iii) Explain the environmental effect of using too much nitrogenous fertilisers. [2]

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8. (a) Fig 8.1 shows an experiment to compare heating efficiencies of fuels A and B.

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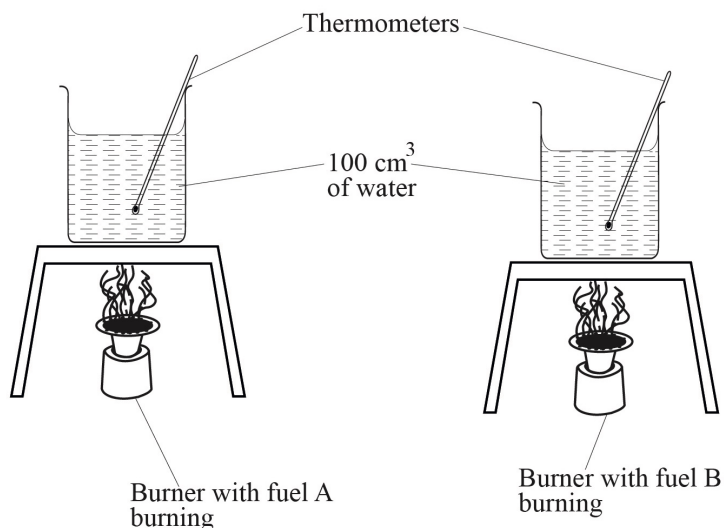


Fig 8.1

Table 8.1 shows temperature changes and amount of fuel used to raise the temperature of 100 cm^3 of water by 20°C .

Table 8.1

fuel type	initial mass of burner + fuel/g	final mass of burner + fuel/g
A	20.0	15.0
B	20.0	18.0

- (i) Explain the term *fuel efficiency*. [2]
- (ii) Identify, with reasons, the most efficient fuel. [2]
- (iii) Suggest
 1. how the experiment can be improved to obtain more accurate results,
 2. why the same temperature rise was used in the experiment.

[3]

- (iv) Explain why the use of bio-fuels is more recommended than the use of fossil fuels. [1]

- (b) Fig. 8.2 shows a method of separating components of crude petroleum oil.

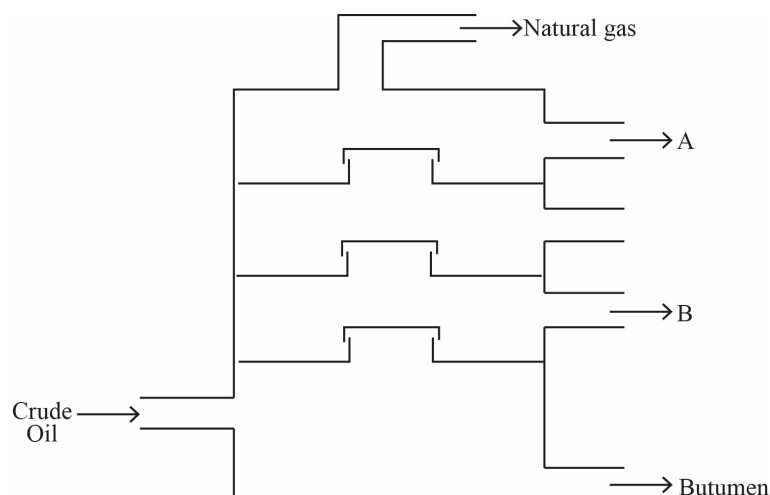


Fig 8.2

Table 8.2 shows some of the components and their respective boiling point ranges.

Table 8.2

fraction	boiling point (°C)
petrol	40 - 75
diesel	250 - 300

- (i) State and explain the method used for separating the components. [2]
- (ii) Suggest, with reasons, the identity of component B. [2]
- (iii) State the main constituent of natural gas. [1]
- (iv) Define the term *cracking*. [2]

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9. (a) Table 9.1 shows different types of alloys, their composition and uses.

Table 9.1

alloy	composition	use
bronze	copper and tin	coins, medals
solder	lead and tin	joining metals
stainless steel	iron, chromium, nickel and carbon	kitchen utensils

Explain why

- (i) the use of alloys is recommended instead of pure metals, [5]
- (ii) bronze is used for making medals,
- (iii) stainless steel is used for making kitchen utensils.
- (b) (i) Water from some sources form crust 'fur' on the walls of the container. [2]
- Describe the source of the fur.
- (ii) Explain how the crust can be removed from the walls of the container. [1]
- (iii) Suggest the effect of the crust on the time it takes for the water to boil. [1]
- (c) Explain why rain water is a safe source of water for domestic use. [2]
- (d) Describe how polyethene is produced from ethane. [4]

10. (a) State any **three**

1. uses of herbs in a community,
2. disadvantages of using herbs.

[6]

(b) Fig. 10.1 shows a separation technique.

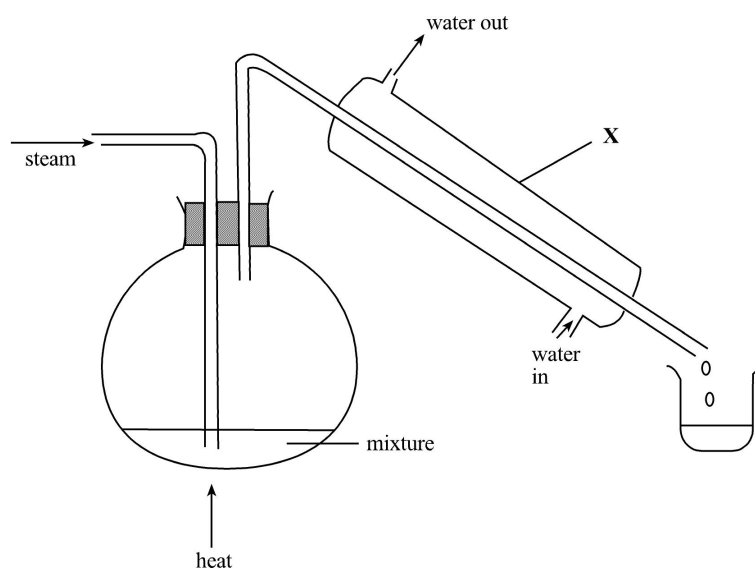


Fig. 10.1

(i) Name

1. the separation technique shown,
2. component X.

[2]

(ii) State the function of X.

[1]

(iii) Describe and explain the principle of this technique.

[2]

(c) Ammonia gas reacts with dilute hydrochloric acid to form ammonium chloride.

(i) Write a balanced chemical equation for the reaction.

[1]

(ii) Name this type of reaction.

[1]

(iii) Calculate the amount of ammonium chloride produced, given that 8.5 g of ammonia was used up.

[2]

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DATA SHEET

The Periodic Table of the Elements

[illegible]

The volume of one mole of any gas is 28 dm^3 at room temperature and pressure (r.t.p.)